HACK FOR SOCIAL CAUSE · EDTECH

Live prototype · Built for Bharat

Student Setu

Bridging Students, Building Futures



studentsetuconnect.netlify.app

COVER PAGE

Hack for Social Cause

PROBLEM STATEMENT / CONCEPT TITLE

Student Setu — AI-powered Curriculum-to-Career Bridge

One free ecosystem from study to job: Class 12 course pathways, skill-gap plans with verified badges, matched jobs, an AI copilot for teachers, counselling and vocational training in Hindi.

THEME

Quality Education & Skill Development

Remote learning · personalised education · empowering teachers · vocational training

TEAM NAME (REGISTERED ON PORTAL)

‹Team Name›

TEAM ID (REGISTERED ON PORTAL)

‹Team ID›

STATE / UT

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Scan for live demo

‹Team Name›

Problem – Proposed Solution

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THE CORE PROBLEM — EDUCATION ISN'T TURNING INTO JOBS

45% of Indian graduates are not job-ready

India Skills Report 2025 (54.81% employable)

~4% of Indians aged 15–59 have formal vocational training

PLFS 2023-24, MoSPI

27:1 students per teacher — little time for prep or personal attention

AISHE 2021-22, Ministry of Education

26% of Web Developer job skills covered by a B.Tech syllabus, even at graduation

Our gap-map analysis (RGPV B.Tech)

PROPOSED SOLUTION — ONE JOURNEY FROM STUDY TO JOB, PLUS TEACHERS

After 12th Pathways

School students

- Class 12 subjects → every course, public & private
- Entrance exams, roadmap, free notes & videos
- Plan B careers if NEET/JEE goes wrong + counselling

Solves: career confusion

Setu Skill Bridge

College students

- Syllabus vs job gap map (45 skills × 13 roles)
- Weekly plan + server-graded, signed badges
- Skills Hub: free AI courses, Prompt Lab, leaderboard

Solves: skills gap

Jobs & Innovation

Graduates & founders

- Update profile → jobs matched to degree & skills
- SIDF: startup events & mentors in govt colleges
- Government startup funding schemes in one place

Solves: unemployment

Teacher Copilot

Teachers

- AI lesson plans, papers & quizzes (English/Hindi)
- Class code join, auto-graded, no student login
- Weak-topic heatmap → one-click re-teach

Solves: overloaded teachers

INNOVATION & UNIQUENESS

- **School → college → job in one place, with Plan B**
- **Tamper-proof badges: answers never reach the browser**
- **Closed teaching loop: quiz → heatmap → re-teach**
- **Bharat-first: Hindi, 2G Lite mode, offline, stay & food map**

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Technical Approach

TECHNOLOGIES — SERVERLESS, LOW-COST STACK

Frontend (PWA) — React 18 · Vite 6 · Tailwind CSS · Framer Motion · React Router · Service Worker · Web Speech API

Serverless API — Netlify Functions (Node 20, TypeScript): copilot · badge · classroom · chat

AI — Google Gemini 2.5 Flash (JSON mode)

Data — Netlify Blobs (class quizzes & results)

Security — HMAC-SHA256 badges, keys on server

METHODOLOGY — HOW IT WORKS

Student journey

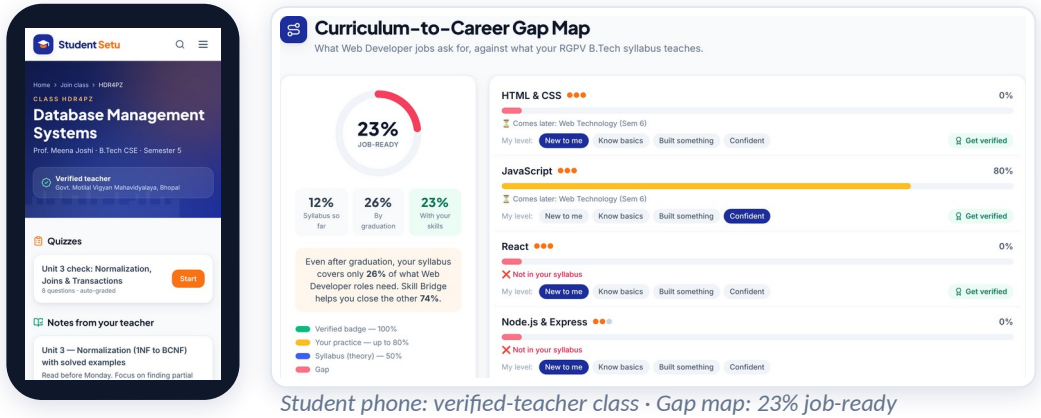
Class 12 subjects → Courses & exams → Gap map + plan → Verified badge → Matched jobs

Teacher loop

Syllabus topic → AI lesson, paper or quiz → Share code on WhatsApp → Students submit, no login → Heatmap → re-teach

Process: agile build–test sprints · Playwright end-to-end tests · GitHub → Netlify auto-deploy

WORKING PROTOTYPE — REAL SCREENS



Student phone: verified-teacher class · Gap map: 23% job-ready

GitHub
prakharporwal518-collab/Student-Setu

Live demo
studentsetuconnect.netlify.app

Demo video
‹Add demo video link›

Extent of AI usage

In product: Gemini writes lesson plans, papers, quizzes, coach tips & chat replies
Guardrails: strict JSON schema, key on server, “review before use”, template fallback
No AI where trust matters: quiz grading & badge signing are deterministic
In development: Claude Code AI assistant for coding & testing, reviewed by the team

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Impact Potential

POTENTIAL IMPACT ON THE TARGET AUDIENCE

School students
Class 9–12

Every course & exam for their subjects · roadmap & free notes · Plan B careers

College students
4.33 Cr in higher education

Exact job-skill gap from Semester 1 · free personal plan · skill proof employers can verify

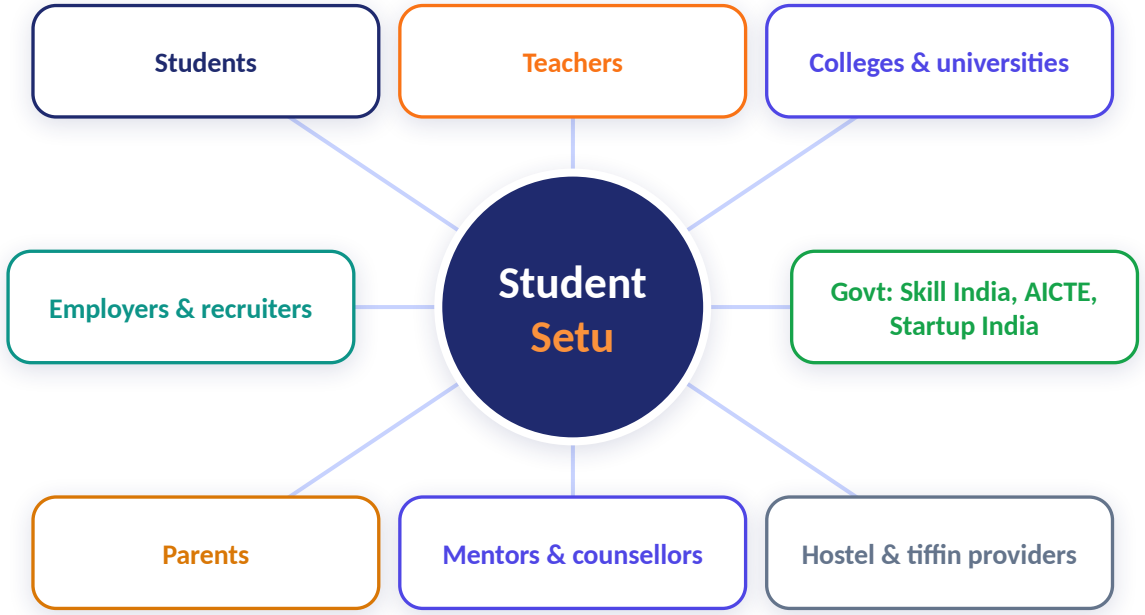
Teachers
15.98 L faculty

Lesson plans & quizzes in minutes · see what the class missed · re-teach with data

Graduates & trade learners
~96% lack formal training

Jobs matched to verified skills · Hindi trade courses · SIDF startup support & funding

STAKEHOLDERS



BENEFITS OF THE SOLUTION

Social
Equal access for rural & Hindi-medium learners; Plan B careers & free counselling ease exam stress

Economic
₹0 for learners, free courses, verified skills → better jobs & wages; saves teachers' prep hours

Educational
Topic-level outcome data; aligned with NEP 2020 — multilingual, vocational & competency-based

Institutional
Colleges see syllabus-vs-industry gaps to update curriculum; employers hire on verified skills

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Feasibility and Risk Analysis

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FEASIBILITY — ALREADY PROVEN BY A LIVE PROTOTYPE

Technical

Live app: 30+ pages, 4 serverless functions; key flows tested end-to-end

Economic

~₹0 at pilot scale on free tiers (Netlify, Gemini); no app-store cost

Operational

Runs in any phone browser; students need no login; English + Hindi

Scalable

Serverless auto-scaling; a new university = one new syllabus file

POTENTIAL CHALLENGES & RISKS → STRATEGIES TO OVERCOME THEM

Risk / challenge	Level	Mitigation strategy
AI gives wrong or low-quality content	High	"Review before use" label · teacher edits all output · strict JSON schema · template fallback
Student exam stress & mental health	High	Plan B careers · human counsellors, not AI advice · Tele-MANAS 14416 helpline shown up front
Poor internet & low-end phones	High	Lite mode (no images or animations) · offline installable PWA · lazy-loaded pages · audio
Fake teacher accounts	Medium	Admin verifies college + employee ID · only verified teachers can run classes
Badge cheating or forgery	Medium	Answers never sent to the browser · server grading · HMAC-SHA256 signature · public verify link
Low teacher adoption	Medium	3 free tries without login · Hindi · WhatsApp sharing · saves prep time from day one
Keeping syllabus data current at scale	Medium	One structured file per university · next: university & teacher uploads, college partnerships
Student data privacy	Low	Students join with class code + name only · no phone or email · API keys only on the server

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Team

TEAM COMPOSITION & ROLES

‹Member 1›
Team Lead · Full-stack

Architecture, serverless functions, deployment

‹Member 2›
Frontend & UI/UX

React screens, Hindi UI, Lite mode, accessibility

‹Member 3›
AI & Backend

Gemini prompts, quiz grading, signed badges

‹Member 4›
Research & Content

Syllabus–skill mapping, vocational modules, testing

CHALLENGES WE SOLVED

- **Keeping test answers secret in a web app**
  → grading moved to the server + signed badges
- **AI must never break a live class**
  → strict JSON output + ready-made templates
- **Students under exam pressure**
  → Plan B careers + counsellors + Tele-MANAS helpline
- **Mapping subjects & syllabus to careers**
  → 24 courses, 22 exam routes, 45 skills × 13 roles
- **Trusting who is really a teacher**
  → admin verification with college & employee ID

OUR JOURNEY

1 **Student-life MVP**

Hostels, tiffin, colleges, local connect

2 **Re-read the problem**

Pivoted to learning-first; life support kept optional

3 **Setu Skill Bridge**

Gap map, personal plan, verified badges

4 **Teacher Copilot**

Lessons, papers, quizzes, class heatmap

5 **Study-to-job**

Pathways, counseling, matched jobs, SIDF funding

6 **Trust & launch**

Teacher verification, Hindi, offline — live

‹Team Name›

References

Data & research

India Skills Report 2025 (CII, Wheebox, AICTE)

taggd.in/industry-reports/isr/india-skills-report-2025

AISHE 2021-22, Ministry of Education

aishe.gov.in

Periodic Labour Force Survey, MoSPI

mospi.gov.in

National Education Policy 2020

education.gov.in — NEP 2020

Government platforms we link to

NTA: JEE Main, NEET UG, CUET UG

nta.ac.in

JoSAA & MCC counselling

josaa.nic.in · mcc.nic.in

Tele-MANAS mental-health helpline

telemanas.mohfw.gov.in

Startup India Seed Fund Scheme

seedfund.startupindia.gov.in

Skill India Digital Hub

skillindiadigital.gov.in

Technology documentation

Google Gemini API

ai.google.dev

Netlify Functions & Blobs

docs.netlify.com

React · Vite · Tailwind CSS

react.dev · vite.dev · tailwindcss.com

Web Speech API & PWAs (MDN)

developer.mozilla.org

Our project

Live demo: studentsetuconnect.netlify.app

Source code: github.com/prakharporwal518-collab/Student-Setu

Demo video: ‹Add demo video link›

Live demo

GitHub